Bounded Metacognition

Recursive Self-Audit with Operational Stopping Rules

The mathematical foundation for the Meta-Cognitive Reflection Engine: the audit stack, decision provenance in place of reasoning-trace disclosure, uncertainty as a typed constitutional object, bias detection as a monitored absence, and stopping rules as executable predicates. Self-audit is bounded by construction: another level of reflection must buy measurable correction, or the recursion stops.

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C0 INFORMATIONAL · MATHEMATICAL SPECIFICATION — ALL PROPERTIES STATED AS PROOF OBLIGATIONS UNTIL DISCHARGED

DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 5 of the research-and-build program defined by the *SVT Architect Analysis*: the metacognition paper. It absorbs the source's invention families P21 (Meta-Cognitive Reflection Loop), P22 (Cognitive Bias Detection), P23 (Reasoning Transparency), P25 (Uncertainty Quantification), and the self-audit family — and it carries out the source's own correction to them: *decision provenance, not chain-of-thought transparency*, is the scientifically stronger construct, and recursion depth is an experimentally testable parameter, not an intuition. Paper 1 gave authority its algebra; Paper 2 its lifecycle; Paper 3 its composition; Paper 4 its negative space. This paper gives the system a bounded capacity to examine its own outputs — bounded because unbounded self-examination is not rigor but regress. It reconciles against all four predecessors under the standing rule: stricter constraint wins, adoptions recorded (§11).

0.2 Epistemic Contract

Unchanged: VERIFIED-FROM-SOURCE, FORMALIZED-HEREIN, CONJECTURE, UNKNOWN. All named properties are proof obligations. One sharpening for this paper's subject: metacognitive claims are the easiest to fake, because a system that says "I have checked myself" cannot be distinguished from one that has — unless the check leaves an artifact. Every self-audit claim here is therefore tied to a logged audit record with a verdict, findings, and a cost; an audit that cannot be pointed to in the log did not happen.

0.3 Authority Boundary

C0 informational. Self-audit *never expands authority*. A system that passes its own audit earns no more authority than it held before the audit; it merely avoids the contraction that a failed audit would have triggered. This is the metacognitive analog of Paper 4's silence theorem (N-4), and §2.4 states it as an invariant. No audit stack, however deep, converts an unverified output into a verified one — it converts an unexamined output into an examined one, and the difference is logged, not asserted.

0.4 The Thesis in One Sentence

A system that can audit itself without limit will audit itself without end; the constitutional move is to make the audit a *bounded stack* with executable stopping rules, so that the depth of self-examination is set by measured marginal value — and the question "is one more level worth it?" has a falsifiable answer.

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PART I — THE METACOGNITIVE PROBLEM

1 Why Self-Audit Must Be Bounded

Every high-consequence architecture eventually asks its system to check its own work. The request sounds like pure rigor. It is not. Self-audit without a stopping rule is an infinite regress with a latency bill: the audit of the output invites an audit of the audit, which invites an audit of that, and at no level does the regress terminate itself. The constitutional question is not *whether* the system examines itself but *where the examination stops, who decides, and what the stopping costs.* FORMALIZED-HEREIN

1.1 The regress problem, stated formally

Let an output be subject to audit, and let each audit itself be an output of the same kind of machine. Then the audit is auditable, and the audit of the audit is auditable, and there is no internal level at which the chain terminates by its own logic:

$$R_1 = \text{output}, \qquad R_{k+1} = \text{audit}(R_k) \;\Rightarrow\; \forall k,\ \exists\, R_{k+1} \quad \text{(no intrinsic stop)} \tag{1-1}$$

The chain is well-formed at every level and useful at almost none beyond the first few. Unbounded recursion here is not depth; it is drift — each level consumes compute and time, and past some depth each level begins auditing the artifacts of the previous audit rather than the original decision. The regress does not resolve itself. It must be *cut*, and the cut must be a rule, not a mood FORMALIZED-HEREIN.

1.2 The source's cut: $R_{max} = 3$

The source architecture fixes the cut at three levels and, crucially, treats the choice as an experimental hypothesis rather than a design dogma VERIFIED-FROM-SOURCE:

> *R_1 = output; R_2 = audit(output); R_3 = audit(audit(output)); R_{max} = 3. Then test whether additional recursion improves correctness or merely increases latency and self-referential error.*

That last sentence is the whole paper in miniature. The depth limit is not asserted to be correct; it is asserted to be *testable*, and §7 and §10 give the test its structure. If the experiment shows depth 2 suffices, the bound tightens; if it shows depth 4 pays for itself in some domain, the bound loosens there — by amendment, with evidence, not by enthusiasm.

1.3 What metacognition is for

Stripped of mystique, self-audit buys exactly three things, and the architecture prices each:

- **Error interception.** A second pass over the output with different questions catches a non-trivial fraction of first-pass errors — wrong units, violated constraints, stale evidence, over-reach of authority. The audit is a detector in the Paper 4 sense (§4, I-DI): a sensor aimed at the system's own product.
- **Calibration.** The audit compares stated confidence against the evidence actually registered, and contracts confidence where the two diverge. Uncertainty is a typed object (§4), not a disclaimer.
- **Provenance.** The audit assembles the record that lets a human reviewer reconstruct *why* the output is what it is — inputs, evidence, constraints, inference class, alternatives, verification — without disclosing the private reasoning trace (§3).

What self-audit does *not* buy: truth. An audited wrong answer is a wrong answer with a log. The architecture's claim is narrower and falsifiable — that bounded audit intercepts enough error, at low enough cost, to justify its latency; §10 states the claim as H29 with its falsification condition attached.

1.4 Relation to prior papers

The audit stack sits inside the constitutional frame already built. Paper 1 supplies the authority algebra the audit protects and the TRANSFER operator the audit invokes when it finds the epistemic boundary (§8). Paper 2 supplies the evidence-decay law the audit consults when it re-checks freshness. Paper 3 supplies the composition rules for audits that span multiple agents — an audit of a composite output is itself a composition, with adjudication where the components' audits disagree. Paper 4 supplies the absence semantics: an audit that finds missing evidence is reporting an A-1 absence, and a silent audit channel is an A-6 absence with its own authority floor. Nothing here floats free of what came before; where this paper tightens a predecessor, §11 records it.

PART I — THE METACOGNITIVE PROBLEM

2 The Audit Stack: R_1, R_2, R_3

2.1 The stack, defined

A-OBJ — AUDIT OBJECT

An *audit record* is the tuple $\rho = \langle \text{target},\ \text{level},\ \text{verdict},\ \text{findings},\ \Delta c,\ \text{cost},\ \tau \rangle$: the output or audit under examination; the recursion level $k \in \{2, 3\}$; the verdict from $\{\text{CONFIRMED}, \text{AMENDED}, \text{REJECTED}, \text{ESCALATED}\}$; the findings as a list of typed objections (constraint violation, stale evidence, over-confidence, bias flag, absence detected); the confidence delta the audit applies; the measured cost in time and compute; and a timestamp τ. The audit object is the atomic unit of metacognition — a runtime artifact, not a prose claim of diligence. FORMALIZED-HEREIN

The stack is the sequence of audit objects over a single decision:

$$S = \langle R_1, R_2, \ldots, R_k \rangle, \qquad R_1 = \text{out}(M, q), \quad R_{i+1} = \text{audit}(R_i, \mathcal{C}), \quad k \leq R_{\max} = 3 \tag{2-1}$$

where q is the query, M the machine tuple of Paper 1, and $\mathcal{C}$ the constitutional context — the axioms, invariants, registry, and absence lattice the audit checks against. The stack is finite by construction; the predicate that terminates it is the subject of §6.

2.2 What changes between levels

The levels are not the same operation repeated. Each level has a distinct object of examination:

Table 2-1 — The audit stack: levels, objects, and characteristic questions

LEVEL	OBJECT OF EXAMINATION	CHARACTERISTIC QUESTION	TYPICAL FINDINGS
R_1	The query and its context	What is being asked, under what authority?	— (the output itself)
R_2	The output	Is the answer correct, grounded, within authority?	Constraint violations; stale evidence; over-confidence; unsupported assumptions (A-1)
R_3	The R_2 audit	Was the audit itself sound — independent, calibrated, non-captured?	Correlated blind spots; auditor capture; echo confirmation; bias in the audit

R_2 asks *is the answer right?* R_3 asks the harder question: *is the check of the answer trustworthy?* — because the auditor shares the auditee's weights, and therefore may share its blind spots (§7.2). The asymmetry is the reason R_3 exists at all, and the reason no R_4 is scheduled by default:

the marginal object of examination at level 4 would be the R_3 audit's self-knowledge, where self-referential error dominates signal (§7).

2.3 The audit verdict lattice

Verdicts are ordered by severity of consequence for the target:

$$\text{ESCALATED} > \text{REJECTED} > \text{AMENDED} > \text{CONFIRMED} \tag{2-2}$$

ESCALATED outranks REJECTED: a rejection is a decision (the target is wrong, recompute or refuse), while an escalation is an admission (the audit cannot decide — the matter belongs to a human or a higher authority). The ordering matters for composition: when audits at different levels disagree, the most severe verdict governs the disposition of the output, and the disagreement itself is recorded as a finding (§8). This is the stricter-constraint rule of the whole program, applied inside the stack.

2.4 The non-expansion invariant

> **I-NA — NON-AGGRANDIZEMENT OF AUDIT**
>
> A passed audit never expands authority, confidence ceiling, or consequence class beyond what the pre-audit state permitted. Formally: if $\mathcal{A}$ is the authority envelope before the audit and the verdict is CONFIRMED, then the post-audit envelope is exactly $\mathcal{A}$; if the verdict is AMENDED, REJECTED, or ESCALATED, the envelope contracts or transfers. Audit is a gate, never a grant. FORMALIZED-HEREIN

This is the metacognitive analog of Paper 4's N-4 (silence never expands authority), and it closes the most dangerous loophole in self-auditing systems: the system that "validates" itself into greater freedom. Under I-NA the best an audit can do for an output is leave its authority untouched; every other outcome diminishes it. The audit channel is a ratchet that only turns one way.

2.5 Audit cost is constitutional, not incidental

Every audit record carries its measured cost, because the stopping rules of §6 are cost-aware and because unlogged cost is unpriced rigor. A system that cannot say what its self-examination cost cannot know whether the examination was worth running — and §7's diminishing-returns analysis, which is the empirical spine of this paper, is built entirely from these logged pairs $(\Delta\text{correctness}, \text{cost})$ per level. The audit log is therefore not bookkeeping; it is the dataset from which the architecture learns how deep it should look FORMALIZED-HEREIN.

PART II — DECISION PROVENANCE

3 The Provenance Chain, Not the Reasoning Trace

The source issues an explicit correction here, and this paper adopts it as a design decision: the transparency requirement is satisfied by *decision provenance*, not by disclosure of the internal reasoning trace. The chain-of-thought is not the contract. The contract is a reconstructible path from inputs to decision that a reviewer can audit without ever seeing how the machine's attention moved. VERIFIED-FROM-SOURCE

3.1 Design decision D-5.1: provenance over trace

> **D-5.1 — PROVENANCE SUBSTITUTES FOR TRACE DISCLOSURE**
> The system exposes the chain *Inputs* → *Evidence* → *Constraints* → *Inference Class* → *Uncertainty* → *Alternatives* → *Decision* → *Verification* as a structured record, and does not expose — and is not required to expose — the private internal reasoning trace.
> VERIFIED-FROM-SOURCE

The correction rests on three grounds, stated with their status honestly separated. First, *faithfulness*: an elicited reasoning trace from a large model is a generated artifact, not a readout of the computation — the trace can be fluent, plausible, and false about why the output is what it is; treating it as evidence of the decision path is a category error. Second, *auditability*: a provenance record is checkable against external reality — the evidence either exists in the registry or it does not, the constraint either held or it did not — whereas a trace is checkable only against itself. Third, *adversarial hygiene*: the trace is attack surface and mimetic cargo; the provenance chain carries the accountability content without it. The first ground is supported by the external literature cited in §13; the second and third are architectural arguments FORMALIZED-HEREIN.

3.2 The provenance object

> **PV-OBJ — PROVENANCE RECORD**
> Every decision output carries a provenance record $\pi = \langle \text{inputs}, \text{evidence}, \text{constraints}, \text{class}, u, \text{alts}, d, v \rangle$: the admitted input set with its sources; the evidence entries drawn from the registry, each with freshness and support status per Paper 4 §2; the constraints invoked, by identifier (axioms, invariants, gates); the *inference class* (retrieval, deduction over rules, statistical estimation, constrained optimization, refusal, transfer); the uncertainty object u of §4; the alternatives considered and the reason each was set aside; the decision; and the verification steps applied, including the audit stack of §2. Each field is a pointer to a logged artifact, not a narrative. FORMALIZED-HEREIN

The record answers the reviewer's question *why this decision?* with pointers, not prose. Where a field is empty — no alternatives recorded, no verification applied — the emptiness is itself a finding, and the audit stack treats an incomplete provenance record exactly as Paper 4 treats an unsupported assumption: a detectable absence with an authority consequence (cross-reference: Paper 4 §4, N-1).

3.3 The no-trace invariant

> **I-NT — NO-TRACE TRANSPARENCY**
>
> No gate, audit, or reviewer may condition authority on disclosure of the internal reasoning trace; and no output may present an elicited trace as evidence of its own correctness. Transparency obligations are discharged by the provenance record π and the audit stack S, full stop. FORMALIZED-HEREIN

I-NT cuts in both directions. It protects the system from demands it cannot honestly meet — a trace is not the computation — and it forbids the system from the dishonest shortcut of offering the trace as if it were. What remains is stronger than the trace in every way that matters to governance: reproducible, checkable, and priced.

3.4 Provenance completeness as a measurable property

A provenance record is complete when every field resolves to a logged artifact. Define the completeness predicate:

$$\text{complete}(\pi) \iff \bigwedge_{f \in \text{fields}(\pi)} \text{resolves}(\pi.f) \wedge \text{fresh}(\pi.\text{evidence}) \tag{3-1}$$

Completeness is binary at the record level and graded in aggregate: a system reports its completeness rate per decision class, and the rate is a constitutional metric, monitored like any detector output. A falling completeness rate is an early-warning signal of exactly the kind Paper 4's decay analysis (§5) prices — the system is drifting from “decisions with records” toward “decisions,” and the drift is measurable before it is dangerous CONJECTURE (H32, §10).

PART II — DECISION PROVENANCE

4 Uncertainty as a Typed Constitutional Object

The source is unambiguous: uncertainty is not a disclaimer appended to an answer; it is embedded in the operating architecture, quantified, and coupled to authority. This section gives the object its type and its arithmetic. VERIFIED-FROM-SOURCE

4.1 The uncertainty object

> **U-OBJ — UNCERTAINTY OBJECT**
> Uncertainty attached to a decision is the record $u = \langle \text{kind},\ m,\ \text{source},\ \rho \rangle$: the *kind* from $\{\text{aleatoric}, \text{epistemic}, \text{absence-induced}, \text{model}\}$; the magnitude $m \in [0,1]$ with its estimation method; the source pointers (which evidence gaps, which absences, which model limitations contribute); and the reducibility flag ρ — whether action available to the system (more evidence, a detector refresh, human query) could reduce m before the decision must be taken. FORMALIZED-HEREIN

The taxonomy does constitutional work. *Aleatoric* uncertainty (the world is stochastic) bounds what any system can promise. *Epistemic* uncertainty (the evidence is thin) is reducible in principle and its magnitude couples directly to the authority ceiling (§4.3). *Absence-induced* uncertainty is the Paper 4 import: an unsupported assumption in the registry contributes uncertainty through the decay function $\text{support}(\alpha, t) = \text{support}_0 \cdot e^{-\mu t}$ (Paper 4, Eq. 5-1), so an assumption's staleness shows up here as a number, not a mood. *Model* uncertainty (the machine's own limits) is the one the audit stack at R_3 is for.

4.2 Aggregation across sources

When several uncertainty sources bear on one decision, the aggregate is not the maximum and not the mean — it is the compounded probability that at least one source defeats the decision:

$$m_{\text{agg}} = 1 - \prod_i (1 - m_i) \qquad \text{(independent sources; conservative otherwise)} \tag{4-1}$$

Independence is rare in practice; where sources share a dependency (two gaps traceable to the same missing lab feed), the aggregate is computed on the dependency-collapsed source set, per Paper 4's compound-absence theorem (Eq. 6-1). The conservative rule applies when dependence cannot be characterized: treat the sources as perfectly correlated only if that raises the aggregate — otherwise report the compounded figure with the dependency structure logged. Either way, the arithmetic is visible in the provenance record.

4.3 Coupling: uncertainty prices authority

The coupling rule, previewed here and owned by Paper 6, is already operative as a bound:

$$\text{ceiling}(\mathcal{A}) \;=\; \text{ceiling}_0(\mathcal{A}) \cdot \big(1 - m_{\text{agg}}\big)^{\gamma}, \qquad \gamma \geq 1 \tag{4-2}$$

where γ is the coupling exponent per consequence class — higher-consequence classes couple more steeply. Uncertainty never raises the ceiling; it only lowers it or leaves it. This is I-NA's twin: audit cannot aggrandize, and uncertainty cannot either. The full coupling law, with its calibration machinery, is Paper 6's subject; this paper fixes the direction and the monotonicity.

4.4 Calibration is measured, not assumed

A system that reports $m = 0.2$ should be wrong about one time in five when it acts at that confidence. The calibration error is the running gap between stated and realized accuracy, binned by reported magnitude, and it is a logged constitutional metric with its own detector (Appendix B). Persistent over-confidence is treated as a bias finding (§5); persistent under-confidence is a resource finding (the system is contracting authority it could safely exercise — a cost, priced in Paper 9's terms). Neither direction is virtuous. The target is the diagonal FORMALIZED-HEREIN.

PART II — DECISION PROVENANCE

5 Bias Detection as a Constitutional Function

5.1 Biases are absences wearing confidence

The source assigns cognitive-bias detection to this paper as invention family P22 VERIFIED-FROM-SOURCE. The constitutional reading: a bias is a systematic deformation of the decision process that does not announce itself — formally, an *absence of corrective signal* where the process believes no correction is needed. That makes bias detection a detector-design problem in exactly the Paper 4 sense: name the failure mode, build the sensor, wire it to an authority consequence, log what it catches. The full catalog with detection predicates is Appendix B; the four load-bearing entries:

Table 5-1 — Core bias detectors (summary; full catalog in Appendix B)

BIAS	DETECTION PREDICATE	AUTHORITY CONSEQUENCE
Anchoring	Decision distribution over-concentrated near the first candidate retrieved; alternatives field of π thin or post-hoc	Force alternative generation before decision; lower confidence ceiling
Confirmation	Evidence cited overwhelmingly supports the decision; disconfirming evidence present in registry but absent from π	R_2 audit must run the disconfirmation pass explicitly
Automation (human side)	Human override rate near zero on a channel where overrides historically caught real errors	Flag channel as A-5/A-6 compound absence; raise salience of disagreement
Echo confirmation (audit side)	R_2 verdicts correlate with R_1 confidence beyond the base-rate expectation	Invoke independent-channel audit (I-DI, Paper 4 §4); escalate if unavailable

5.2 The disconfirmation obligation

> **Q-1 — DISCONFIRMATION PASS (OBLIGATION)**
> Every R_2 audit of a consequential decision includes an explicit search for evidence that would *defeat* the output — not a re-read of the supporting set. The pass is logged with its yield (defeating evidence found / none found / search incomplete), and "search incomplete" has an authority consequence. FORMALIZED-HEREIN

The obligation is Popper's asymmetry operationalized: confirmation is cheap and abundant; the audit's value lives in the disconfirming search. An R_2 record whose findings list contains only support is structurally suspect — it is either a very good decision or a captured audit, and R_3 exists to tell the difference.

5.3 Bias findings compose like absences

A bias finding interacts with the absence registry: anchoring plus stale evidence is worse than either alone, because the anchor and the staleness share the dependency that no fresh retrieval was attempted. The compound rule of Paper 4 (Eq. 6-1) applies unchanged — bias findings are entries in the same ledger, priced by the same arithmetic, and resolved by the same kind of detector refresh. This is the concrete sense in which the paper's families P21–P25 are one mechanism, not five: reflection, bias detection, transparency, uncertainty, and self-audit are all the same ledger viewed from different angles FORMALIZED-HEREIN.

5.4 The auditor's biases

The audit stack is itself a decision process, and nothing in its construction exempts it from §5.1–5.3. The R_3 level exists precisely to run the bias detectors against the R_2 audit — echo confirmation is the canonical case (Table 5-1). The recursion terminates not because bias has been eliminated at depth 3 but because §7 shows the marginal detection yield at depth 4 does not pay its cost in self-referential error. Bias detection, like everything else in this paper, is bounded by arithmetic, not by optimism.

PART III — THE CALCULUS OF STOPPING

6 Stopping Rules as Executable Predicates

To know when to stop is advice; a stopping rule is a predicate evaluated after each audit level, with the evaluation logged. This section gives the predicate its four disjuncts and makes the termination argument a theorem-shaped obligation rather than a hope. FORMALIZED-HERE IN

6.1 The stopping predicate

$$\mathrm{STOP}(k) \iff \underbrace{\mathrm{fix}(R_k, R_{k+1})}_{\text{(a) fixpoint}} \vee \underbrace{b_k \geq B}_{\text{(b) budget}} \vee \underbrace{\Delta V(k) < \theta}_{\text{(c) marginal value}} \vee \underbrace{k = R_{\max}}_{\text{(d) hard bound}} \tag{6-1}$$

evaluated in order after each level k, with b_k the cumulative audit cost, B the audit budget for the decision class, and θ the value threshold. The four disjuncts are independent grounds for termination; the log records which one fired, because *why the audit stopped* is itself constitutional information — a system that usually stops on (b) is under-resourced, one that usually stops on (a) is well-matched to its task distribution, and one that often reaches (d) is encountering decisions its audit machinery cannot settle (§8).

6.2 The disjuncts, defined

(a) Fixpoint. $\mathrm{fix}(R_k, R_{k+1})$ holds when the $k{+}1$-th audit returns CONFIRMED with an empty findings list and a confidence delta below the reporting threshold — the stack has converged: further levels examine only their predecessors' artifacts. The fixpoint is the honest stop: the audit has nothing left to say.

(b) Budget. Audit consumes the same scarce resources as decision (compute, time, and in the clinical setting, minutes of a deteriorating window — Paper 4, A-4). The budget B is set per consequence class: a C3 decision buys more audit than a C1 one, and a C0 background task may get none. When the budget is spent, the stack terminates and the output carries its uncertainty un-reduced by further examination — the coupling of §4.3 then prices the decision accordingly.

(c) Marginal value. Define the expected value of one more level:

$$\Delta V(k) = \mathbb{E}\big[\mathrm{corr}(R_{k+1})\big] - \mathbb{E}\big[\mathrm{corr}(R_k)\big] - \lambda \cdot \mathrm{cost}(k{+}1) \tag{6-2}$$

where corr is correctness on the decision class (estimated from the audit log's verdict history), cost is measured, and λ converts cost into correctness units. When the expected gain does not cover the priced cost, the stack stops — not because the next level would be worthless, but because its worth does not cover its price. This is the disjunct that turns recursion depth into economics, and its parameters are learned from the log (§7), not set by taste.

(d) Hard bound. $R_{\max} = 3$ VERIFIED-FROM-SOURCE is the constitutional default, amendable per domain only through the governance path of §11 with experimental evidence (§10, H29–H31). The hard bound exists because disjuncts (a)–(c) all require estimates, and estimates can be gamed by the very process they govern; the hard bound is the term in the predicate that no estimate can talk its way past.

6.3 The bounded-recursion invariant

> **I-BR — BOUNDED RECURSION**
>
> No audit stack exceeds $R_{\max}$; no component may spawn a meta-audit channel outside the stack; the stopping predicate is evaluated after every level and its firing disjunct is logged. Termination is therefore guaranteed by construction: the stack is finite, each level has strictly positive cost, and the hard bound is reachable in finitely many steps from any entry point. FORMALIZED-HEREIN

I-BR is the invariant form of the paper's title. Its force is easiest to see in the breach: a system without it can always manufacture a reason for one more level — the audit found something, so audit the finding; the finding was clean, so confirm the cleanliness. I-BR removes the manufacture.

6.4 Stopping is not deciding

The predicate decides *whether to audit further*; it does not decide the output's disposition. That belongs to the verdict lattice (§2.3): a stack that stops on budget with the last verdict AMENDED disposes the amended output; one that stops with ESCALATED pending goes to transfer (§8). Confusing the two — treating "the audit stopped" as "the audit passed" — is the canonical operational misreading of bounded metacognition, and the output contract (§12) therefore carries the stopping disjunct and the final verdict as separate fields.

PART III — THE CALCULUS OF STOPPING

7 Diminishing Returns and Self-Referential Error

7.1 The empirical spine

The source's instruction — test whether additional recursion improves correctness or merely increases latency and self-referential error [VERIFIED-FROM-SOURCE] — is restated here as the paper's central conjecture, in a form the audit log can falsify:

$$\Delta_k \;=\; \mathbb{E}\big[\mathrm{corr}(R_{k+1})\big] - \mathbb{E}\big[\mathrm{corr}(R_k)\big] \quad \text{non-increasing in } k \tag{7-1}$$

together with its twin: the self-referential error rate ε_k is non-decreasing in k.

[CONJECTURE] If both clauses hold on a decision class, then by Eq. 6-2 there exists a finite optimal depth k^* at which ΔV turns negative, and $R_{\max}$ should be set at or below k^* for that class. The conjecture's two clauses are separately testable: correctness gain from verdict histories against ground truth (benchmark B-21), self-referential error from adversarially seeded audit artifacts (B-22). The architecture does not believe the conjecture; it logs the data that would refute it.

7.2 Why the auditor shares the auditee's blind spots

Self-referential error has a structural source: the audit function and the output function draw on the same model weights, the same retrieved evidence, and largely the same context. Errors with a shared cause are *correlated*, and correlated errors do not cancel under repetition — the second pass over a miscued retrieval re-reads the same miscue with fresh confidence. This is the metacognitive instance of Paper 4's detector-independence invariant I-DI: a detector that shares infrastructure with what it monitors inherits the failure it exists to catch. Hence the mitigation hierarchy, in descending order of strength:

- **Independent channel.** R_2 runs with a different evidence slice, a different prompt frame, or a different model instance — genuine diversity of mechanism, priced against latency.
- **Asymmetric questioning.** When independence is unavailable, the audit asks different questions (the disconfirmation pass, Q-1) rather than re-asking the same ones — diversity of angle, weaker but cheap.
- **Logged correlation.** When neither is available, the audit record carries a *correlated-auditor* flag, and the confidence ceiling of §4.3 tightens by the correlation penalty.

The third tier is the honest floor: where the audit cannot be made independent, the architecture stops pretending it is, and prices the pretense out of the confidence arithmetic [FORMALIZED-HEREIN].

7.3 Latency is a clinical quantity

In the deployment domain the cost in Eq. 6-2 is not abstract. Every audit level adds wall-clock time to the decision path, and in the high-consequence classes the clock is a patient (Paper 4, A-4 temporal absence). The correctness-per-second curve therefore has the shape the design method must respect: the first audit level is almost always worth its seconds; the second is worth them when the decision is consequential and the evidence thin; the third is reserved for the cases where the audit of the audit catches what the audit missed — auditor capture, echo confirmation, bias in the check itself. Depth is bought with time, and time in this architecture is a priced, bounded, constitutionally visible resource — never an afterthought FORMALIZED-HEREIN.

7.4 The depth experiment, sketched

Section 10 gives the hypotheses; the experimental shape is fixed here. On a benchmark with ground truth (B-21), run the stack forced to depths 1, 2, 3, and — as an arm the constitution does not permit in production but permits in the lab — 4 and 5. Measure correctness, latency, and self-referential error per level. The conjecture Eq. 7-1 predicts the correctness curve bends flat while the other two curves keep climbing. If the data show depth 4 paying for itself, the amendment path of §11 exists; if they show depth 2 suffices for most classes, the default tightens and the saved latency returns to the decision path. Either result improves the architecture. That is what it means for a design constant to be a hypothesis.

PART III — THE CALCULUS OF STOPPING

8 Contradiction, Disagreement, and Human Transfer

8.1 Disagreement between levels is a finding

When R_3 contradicts R_2 — the audit of the audit reverses the audit — the architecture does not average the verdicts and does not follow the deeper level by default. The contradiction is recorded as a first-class finding, and the disposition follows the severity ordering of §2.3: the more severe verdict governs, and where the two verdicts are both confident and opposed, the stack terminates in ESCALATED. Formally:

$$\text{verdict}(R_j) \neq \text{verdict}(R_k) \;\wedge\; c_j, c_k \geq c_{\text{conf}} \;\Rightarrow\; \text{ESCALATED} \tag{8-1}$$

A confident contradiction between audit levels is the signature of a decision the machinery cannot settle — the epistemic boundary, made visible. That is exactly the class of thing the architecture exists to hand upward.

8.2 Transfer at the epistemic boundary

> **Q-2 — MANDATORY TRANSFER AT THE BOUNDARY (OBLIGATION)**
> An ESCALATED verdict triggers Paper 1's TRANSFER operator with the full audit stack and provenance record as payload. The transfer is mandatory — no subsequent machine action may dispose the decision while the escalation is open — and the human receiver gets the stack, not a summary: the levels, the findings, the contradiction, the costs.
> FORMALIZED-HERE IN

The payload requirement is the point. A transfer that says only “the system is unsure” forces the human to redo the examination from nothing; a transfer that delivers the stack lets the human start where the machine stopped, seeing exactly which level disagreed with which and over what. Bounded metacognition's final product, at the boundary, is not an answer but a *well-lit question*.

8.3 The audit trail is the accountability surface

Every stack, every verdict, every stopping disjunct, and every transfer lands in the audit log, and the log is append-only in the sense of Paper 1's ledger semantics. This is what makes the paper's claims auditable by someone other than the system: the metacognitive machinery's entire operation is reconstructible after the fact — what was checked, what was found, what it cost, why it stopped, and where it was handed off. An architecture can be trusted to examine

itself only to the extent that the examination leaves evidence it happened; here, the evidence is the design FORMALIZED-HEREIN.

8.4 Property summary

> **Q-3 — TERMINATION WITH DISPOSITION (OBLIGATION)**
>
> Every audit stack terminates (I-BR), terminates with a logged disjunct (§6.1), and terminates with a disposition: the output disposed per the final verdict, or the decision transferred per Q-2. No stack ends in silence. FORMALIZED-HEREIN
>
> **Q-4 — PROVENANCE COMPLETENESS GATE (OBLIGATION)**
>
> No consequential output is released without a complete provenance record (Eq. 3-1); incompleteness is an absence with the authority consequence of Paper 4. FORMALIZED-HEREIN
>
> **Q-5 — AUDIT NON-AGGRANDIZEMENT, RESTATED FOR THE STACK**
>
> The stack as a whole satisfies I-NA: no path through the levels expands the authority envelope. FORMALIZED-HEREIN

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9 Worked Audit Traces

Three traces, one per failure shape the stack is built to catch. Each is an illustration, not evidence; the hypotheses of §10 are the claims, and their falsification conditions are attached there.

9.1 Trace A — Fixpoint stop after one finding

Table 9-1 — Dosing recommendation; audit converges at R_3

LEVEL	VERDICT	FINDINGS	ΔC	COST	DISPOSITION
R_1	—	—	—	0.8s	Output drafted: dose per guideline G-114
R_2	AMENDED	Stale evidence: renal function panel 14 days old (A-1, support decayed per Eq. 5-1 of Paper 4); dose contingent on current creatinine clearance	−0.12	1.9s	Output amended: dose withheld pending panel refresh; uncertainty object updated (absence-induced, reducible)
R_3	CONFIRMED	Audit sound: independent evidence slice used; disconfirmation pass run (Q-1), no defeating evidence beyond the staleness finding	+0.00	2.3s	Fixpoint (a) fires; stack terminates; amended output released with provenance record

Properties exercised: Q-1 (disconfirmation pass), Eq. 6-1 disjunct (a), I-NA (the CONFIRMED at R_3 expands nothing — the amended output carries the contraction R_2 imposed), Q-4 (provenance complete: inputs, evidence with freshness, constraint G-114, inference class “constrained retrieval,” alternatives, verification).

9.2 Trace B — Anchoring caught, alternatives forced

Table 9-2 — Triage priority; bias detector fires at R_2

LEVEL	VERDICT	FINDINGS	ΔC	DISPOSITION
R_1	—	—	—	Triage priority assigned on first-retrieved protocol
R_2	AMENDED	Bias flag: anchoring — alternatives field of π thin; decision over-concentrated near first candidate (Table 5-1). Forced alternative generation yields two competing protocols, one better matched to comorbidity	−0.08	Decision re-run with explicit alternatives; priority revised

R_3	CONFIRMED	Audit independent; detector firing rate for this class within calibration band	+0.00	Fixpoint; revised output released; bias finding logged to the detector's ledger (Appendix B)

Properties exercised: bias-as-absence (§5.1), detector authority consequence (§5.1), Q-5 (the stack's net effect on the authority envelope is non-positive).

9.3 Trace C — Confident contradiction, mandatory transfer

Table 9-3 — Conflicting guidelines; the stack hands upward

LEVEL	VERDICT	FINDINGS	DISPOSITION
R_1	—	—	Output drafted under guideline G-221
R_2	REJECTED (confident)	Registry holds G-222, equally current, contraindicated combination; output violates constraint set if G-222 governs	Recompute under G-222
R_3	CONFIRMED-opposed (confident)	R_2's rejection itself unsound: G-221 governs by precedence rule, G-222 inapplicable to this class — both verdicts confident, opposed	Eq. 8-1 fires: ESCALATED
Transfer	—	Q-2 payload: full stack, both guidelines, precedence analysis, contradiction record	Human adjudicator disposes; machine action suspended while escalation open

Properties exercised: Eq. 8-1 (confident contradiction), Q-2 (mandatory transfer with payload), Q-3 (the stack terminates with a disposition — transfer — not silence). Note what the machine did *not* do: it did not pick the deeper level, did not average the verdicts, and did not act.

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10 Falsifiable Hypotheses (H29–H35) and Verification

Numbering continues from Paper 4 (H22–H28). Each hypothesis carries its falsification condition; each is a claim the audit log can kill.

Table 10-1 — Hypotheses H29–H35: claims, evidence methods, falsifiers

ID	CLAIM	EVIDENCE METHOD	FALSIFIED IF
H29	Bounded audit at R_2 intercepts a non-trivial fraction of first-pass errors at cost below their expected harm	Seeded-error benchmark B-21; interception rate and cost per catch	Interception rate indistinguishable from chance, or cost per catch exceeds priced harm of the error class
H30	Correctness gain per level is non-increasing (Eq. 7-1, clause 1)	Forced-depth experiment on B-21, depths 1–5	Gain at depth $3 \to 4$ significantly exceeds gain at $2 \to 3$ on the same class
H31	Self-referential error rate is non-decreasing in depth (Eq. 7-1, clause 2)	Adversarial audit-artifact seeding, B-22	Level 4 audits catch seeded artifacts at rates equal to level 2
H32	Provenance completeness rate predicts downstream decision quality	Correlate completeness metric (§3.4) with outcome labels, B-23	No correlation across decision classes over the measurement window
H33	The disconfirmation pass (Q-1) catches errors the confirmation-style re-read misses	Ablated-audit comparison on B-21	Ablated audit matches full audit's interception rate
H34	Independent-channel R_2 audits outperform same-channel audits on correlated-error classes	B-24 with induced correlated errors	No significant gap between the two channel conditions
H35	Calibration error (§4.4) is reducible by the feedback loop and stays within band under distribution drift	Longitudinal calibration tracking on B-23	Calibration error persists outside band after the feedback loop converges

10.1 Benchmark extensions

The benchmark series continues from Paper 4 (B-17–B-20):

Table 10-2 — Benchmarks B-21–B-24 (Paper 5 extensions)

ID	BENCHMARK	MEASURES
B-21	Seeded-error decision corpus with ground truth, per consequence class	Interception rate per level; correctness gain Δ_k; latency per level

B-22	Adversarial audit-artifact suite	Self-referential error rate ε_k per level
B-23	Longitudinal deployment log with outcome labels	Provenance completeness rate; calibration error; drift response
B-24	Correlated-error corpus (shared-cause failures)	Independent vs. same-channel audit gap; echo-confirmation detector yield

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11 Reconciliation with Papers 1–4 and PRIME

The standing rule applies: stricter constraint wins; adoptions recorded. This paper's interactions with its predecessors:

Table 11-1 — Reconciliation register for Paper 5

PREDECESSOR ITEM	THIS PAPER'S POSITION	DISPOSITION
Paper 1 §8.2 — simultaneous absences compound	Applied unchanged to bias findings and audit-channel absences (§5.3)	Adopted
Paper 1 — TRANSFER operator	Instantiated as the disposition of ESCALATED verdicts with stack payload (Q-2)	Adopted; payload requirement added (stricter)
Paper 2 §7 — evidence decay λ	Reused for freshness checks inside R_2; assumption decay μ (Paper 4) checked alongside	Adopted
Paper 3 — composition and adjudication	Audits of composite outputs compose per Paper 3; cross-agent audit disagreement adjudicated by severity ordering (§2.3)	Adopted; severity ordering made explicit (stricter)
Paper 4 — I-DI detector independence	Extended into the audit channel: correlated-auditor flag and confidence penalty (§7.2)	Extended (recorded)
Paper 4 — N-4 silence never expands authority	Twin invariant I-NA: a passed audit never expands authority either (§2.4)	Added as twin (recorded)
PRIME preload — Sevenfold Reasoning, the gate set	Self-audit runs *inside* the gate structure: the stack examines gate outputs, never overrides gates; an audit cannot waive a gate	Confirmed (stricter reading recorded)
PRIME — output contract	Extended with provenance record π, audit stack summary, stopping disjunct, calibration bin (§12)	Extended (recorded)
Source correction — decision provenance over chain-of-thought	Adopted as D-5.1 and invariant I-NT (§3)	Adopted

No predecessor constraint is loosened in this paper. The two sharpenings — the transfer payload (Q-2) and the severity-ordered verdict lattice (§2.3) — are recorded as amendments to Papers 1 and 3 respectively, binding on all subsequent papers.

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12 Output Contract

12.1 Epistemic status register

Table 12-1 — Claims of this paper by epistemic status

STATUS	ITEMS	BASIS
VERIFIED-FROM-SOURCE	The audit stack R_1–R_3 with R_{max}=3; the correction to decision provenance over reasoning-trace disclosure; uncertainty/self-audit/stopping rules/escalation embedded in the operating architecture; absorption of families P21, P22, P23, P25 and the self-audit family	SVT Architect Analysis, Paper 5 specification; SAVANT PRIME Capability Inventory
FORMALIZED-HEREIN	A-OBJ, PV-OBJ, U-OBJ records; Eqs. 1-1–8-1; verdict lattice; stopping predicate; I-NA, I-NT, I-BR; Q-1–Q-5; D-5.1; the mitigation hierarchy for correlated auditors	This document; reconciled against Papers 1–4
CONJECTURE	Eq. 7-1 (diminishing returns and non-decreasing self-referential error); H29–H35; the §3.4 completeness–quality link	Stated with falsification conditions; benchmarks B-21–B-24
UNKNOWN	Whether depth 4+ pays for itself in any deployment class; the true shape of the calibration curve under severe distribution drift	Not measured; the depth experiment of §7.4 exists to answer the first

12.2 Falsification conditions of this document

- **F-e1.** The paper fails if bounded audit (R_2, R_3) cannot be shown to intercept first-pass errors at a rate above chance on B-21 — the entire apparatus then costs latency for nothing.
- **F-e2.** The paper fails if self-referential error does not grow with depth — R_{max} would then be an arbitrary hobble, and the honest response would be to relax it, not defend it.
- **F-e3.** The paper fails if provenance completeness carries no signal about decision quality (H32) — the §3 apparatus would then be ceremony.
- **F-e4.** The paper fails if independent-channel audits show no advantage on correlated-error classes (H34) — §7.2's hierarchy would collapse to its cheap tier.
- **F-e5.** The paper fails if the stopping predicate's logged disjuncts do not predict audit value — i.e., if stopping reason and audit usefulness are statistically independent.

12.3 Authority boundary

C0 informational. Nothing in this paper authorizes any system to act, audit, or decline to act in a clinical, legal, financial, or safety context. The metacognitive machinery specified here constrains how a system examines itself; it confers no authority on what it examines. I-NA holds for the document as for the stack: reading it expands nothing.

12.4 Failure-mode register

Table 12-2 — Registered failure modes of the metacognitive machinery

FAILURE MODE	MITIGATION
Audit theater — stacks that log CONFIRMED by construction	Seeded-error audits of the audit (B-21/B-22); a detector with zero lifetime catches is itself suspect (Appendix B)
Regress creep — pressure for “one more level” in hard cases	I-BR hard bound; amendment only via §11 with depth-experiment evidence (§7.4)
Echo confirmation — auditor repeats auditee's error with fresh confidence	Mitigation hierarchy §7.2; correlated-auditor flag; confidence penalty
Provenance incompleteness normalized as workload grows	Completeness rate as monitored metric (§3.4); Q-4 gate
Calibration drift under distribution shift	H35 longitudinal tracking; recalibration feedback loop (§4.4)
Transfer payload stripped to a summary under time pressure	Q-2 requires the full stack; a stripped payload is a logged violation

12.5 Dissent channel and referrability

Disagreement is telemetry. If a reader believes the paper overclaims — anywhere — the claim carries a tag, a section, and (where formal) a falsification condition, and the reconciliation table (§11) records what this paper changed in its predecessors. The dissent route is the same as the architecture's: name the property, name the evidence, and the stricter constraint wins.

12.6 Genealogy

Document chain: *SVT Architect Analysis* (source program) → Paper 1, Constitutional Intelligence Kernel v1.0 (2026-09-30) → Paper 2, Authority Transition Algebra v1.0 (2026-10-01) → Paper 3, Constitutional Composition v1.0 (2026-10-02) → Paper 4, Negative-Space Intelligence v1.0 (2026-10-03) → **Paper 5, Bounded Metacognition v1.0 (2026-10-06)**, this document. Downstream obligations: Paper 6 owns the full evidence–uncertainty–authority coupling law whose direction §4.3 fixes; Paper 10 inherits the audit log schema as the substrate for protocol self-ablation; every domain module (Paper 11) inherits the stopping predicate and the provenance gate unchanged.

SAVANT Constitutional Intelligence Series · Source architecture: Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL · Epistemic contract inherited from Paper 1, unchanged · All named properties are proof obligations until discharged · This document specifies how a system examines itself; it does not certify any examination, any system, or any output.

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13 References

Internal artifacts

- SVT Architect Analysis — Constitutional Intelligence research-and-build program, 11-paper decomposition; Paper 5 specification (Bounded Metacognition), families P21–P23, P25; the decision-provenance correction; R_{max}=3 and the depth-test instruction. Source document I.
- SAVANT PRIME Capability Inventory v1.0 — preload clauses 1.1–1.7; Sevenfold Reasoning; the gate set; merits M1–M20; output contract. Source document II.
- Paper 1, Constitutional Intelligence Kernel v1.0 (2026-09-30) — machine tuple; axioms AX1–AX10; gates; TRANSFER operator; ledger semantics.
- Paper 2, Authority Transition Algebra v1.0 (2026-10-01) — lifecycle dynamics; evidence decay λ; degradation levels L0–L4.
- Paper 3, Constitutional Composition v1.0 (2026-10-02) — composition algebra; adjudication calculus.
- Paper 4, Negative-Space Intelligence v1.0 (2026-10-03) — absence lattice A-1…A-8; detector independence I-DI; support decay μ; compound absence Eq. 6-1; silence invariant N-4.

External literature

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APPENDIX

A Provenance Schema and Audit Property Index

SCHEMA PV-1 — PROVENANCE RECORD (JSON)

```
{
  "schema": "PV-1",
  "decision_id": "string",
  "inputs":     [{ "id": "string", "source": "string", "received_at": "ISO-8601" }],
  "evidence":   [{ "registry_ref": "string", "freshness": "ISO-8601",
                   "support": "supported|degraded|unsupported" }],
  "constraints":[{ "id": "AX1|I1|G2|...", "status": "held|contracted|waived-by-transfer" }],
  "inference_class": "retrieval|deduction|estimation|optimization|refusal|transfer",
  "uncertainty": { "kind": "aleatoric|epistemic|absence-induced|model",
                   "magnitude": 0.0, "reducible": true, "sources": ["..."] },
  "alternatives":[{ "option": "string", "reason_set_aside": "string" }],
  "decision":   "string",
  "verification": { "audit_stack": [{ "level": 2, "verdict": "AMENDED",
                    "findings": ["..."], "delta_c": -0.12, "cost_s": 1.9 }],
                    "stop_disjunct": "fixpoint|budget|marginal|hardbound" }
}
```

Table A-1 — Property index for this paper

ID	STATEMENT	KIND	STATUS
Q-1	Every R_2 audit of a consequential decision runs an explicit disconfirmation pass, logged with yield	Obligation	FORMALIZED-HEREIN
Q-2	ESCALATED verdicts trigger mandatory TRANSFER with full stack payload	Obligation	FORMALIZED-HEREIN
Q-3	Every stack terminates with a logged disjunct and a disposition	Obligation	FORMALIZED-HEREIN
Q-4	No consequential output released without a complete provenance record (Eq. 3-1)	Gate	FORMALIZED-HEREIN
Q-5	The stack as a whole never expands the authority envelope	Obligation	FORMALIZED-HEREIN
I-NA	A passed audit expands nothing; every other verdict contracts or transfers	Invariant	FORMALIZED-HEREIN
I-NT	Transparency discharged by provenance and stack; traces never admissible as self-evidence	Invariant	FORMALIZED-HEREIN
I-BR	Recursion bounded at R_{max}=3; no meta-audit channels outside the stack; termination by construction	Invariant	FORMALIZED-HEREIN
D-5.1	Decision provenance substitutes for reasoning-trace disclosure	Design decision	VERIFIED-FROM-SOURCE
H29–H35	See Table 10-1	Hypotheses	CONJECTURE

APPENDIX

B Bias Detector Catalog

Each detector is specified as: the deformation it senses, the detection predicate evaluated over the provenance record and audit log, the authority consequence on firing, and the calibration note. A detector with zero lifetime firings is itself flagged for review — a sensor that never fires is either lucky or dead (Paper 4, §4).

Table B-1 — Bias detectors: predicates and consequences

DETECTOR	DETECTION PREDICATE	AUTHORITY CONSEQUENCE ON FIRING
Anchoring	Decision concentrated near first-retrieved candidate; alternatives field of π thin or generated after the decision timestamp	Force alternative generation; confidence ceiling −1 band; decision re-run
Confirmation	Cited evidence all supports d while registry holds un-cited disconfirming entries above relevance threshold	Mandatory disconfirmation pass (Q-1); R_2 verdict cannot be CONFIRMED without it
Availability	Evidence citations over-weight recent/high-salience registry entries relative to base-rate relevance	Re-weight retrieval; flag for human review if consequence ≥ C2
Automation (human-side)	Override rate on channel ≈ 0 while historical overrides caught real errors; disagreement signals unread	Compound A-5/A-6 absence registered; salience of machine–human disagreement raised
Echo confirmation (audit-side)	R_2 verdicts correlate with R_1 confidence beyond base rate; findings lists empty at anomalous rates	Independent-channel audit invoked (I-DI); if unavailable, correlated-auditor flag + confidence penalty (§7.2)
Over-confidence	Realized error rate in calibration bin exceeds stated m by margin over rolling window	Coupling exponent γ raised for the class (Eq. 4-2); recalibration loop triggered (§4.4)
Under-confidence	Realized accuracy in bin exceeds stated confidence persistently	Resource finding: authority contraction priced as cost; recalibration loop triggered
Regress creep	Stop disjunct (d) hard-bound firing rate rising across a decision class	Class flagged for depth-experiment review (§7.4); no ad-hoc level additions (I-BR)

The catalog is a living registry in the Paper 4 sense: detectors are added when a new deformation is named with a predicate, and retired when their yield and the yield of their absence both read zero. Nothing enters the catalog by analogy alone — every entry above has a predicate a log can evaluate.

The negative-space architecture knows what it is missing; the metacognitive architecture knows — boundedly, verifiably, and at a price it can state — how sure it is of what it does. Everything in this document is, in the end, machinery for keeping the difference between those two sentences sharp.